

CodeBuddy Local

Offline AI error explainer and fix assistant for Snapdragon PCs



The problem



Unreliable internet

Many students code where connectivity is patchy, so cloud AI helpers fail when they are needed.



Privacy and cost

Pasting coursework or project code into cloud tools raises privacy concerns, and free tiers run out.



Cryptic errors

Beginners lose hours to stack traces they cannot read, and search results rarely match their exact code.

Our solution

1

Paste

Code snippet and error message into a simple local web UI

2

Analyze

A small code LLM runs on-device on the Snapdragon NPU

3

Understand

Plain-English explanation and a suggested fix, with a beginner mode

100% offline. No code ever leaves the laptop.

Built for Snapdragon



On-device AI

- Small code LLM (Qwen2.5-Coder or Phi class)
- Models from Qualcomm AI Hub / open source
- ONNX Runtime with QNN for NPU acceleration



Stack and design

- Python backend, lightweight web front end
- Runs fully offline after setup
- Low power use keeps laptops on battery longer

NPU vs CPU results

Metric	CPU	Snapdragon NPU
Time to first token	___ ms	___ ms
Tokens per second	___	___
Total time per fix	___ s	___ s

Measured on a Snapdragon-powered HP PC using the same prompt and model.

Impact and next steps

Free

No subscription or API keys

Private

Code stays on the device

Offline

Works with no internet

Future scope

- VS Code extension for one-click fixes
- Support for more languages and test-failure explanations